

where F is the force on each particle, m_1 and m_2 are their masses, r is the distance between them, and G is a gravitational constant. Assuming that m_1 and m_2 are constant, we obtain

$$\lim_{r \rightarrow \infty} F = \lim_{r \rightarrow \infty} G \frac{m_1 m_2}{r^2} = 0.$$

This tells us that as the distance between the particles increases without bound, the force of attraction approaches 0. Theoretically, there is always *some* attraction; however, if r is very large, the attraction cannot be measured with conventional laboratory equipment.

We shall conclude this section by stating a formal definition of $\lim_{x \rightarrow a} f(x) = \infty$. The main difference from our work in Section 2.2 is that instead of showing that $|f(x) - L| < \epsilon$ whenever x is near a , we consider any (large) positive number M and show that $f(x) > M$ whenever x is near a .

Definition (2.19)

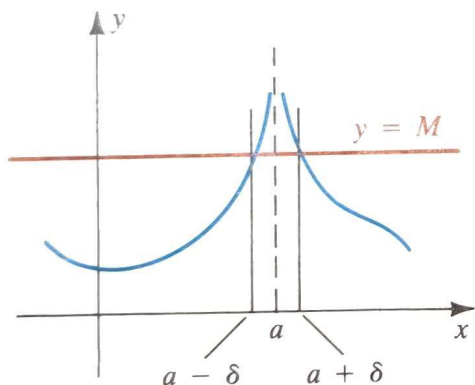
$$\lim_{x \rightarrow a} f(x) = \infty$$

means that for every $M > 0$, there is a $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta, \text{ then } f(x) > M.$$

FIGURE 2.34

$$\lim_{x \rightarrow a} f(x) = \infty$$



For a graphical interpretation of Definition (2.19), consider any horizontal line $y = M$, as in Figure 2.34. If $\lim_{x \rightarrow a} f(x) = \infty$, then whenever x is in a suitable interval $(a - \delta, a + \delta)$ and $x \neq a$, the points on the graph of f lie *above* the horizontal line.

To define $\lim_{x \rightarrow a} f(x) = -\infty$ we may alter Definition (2.19), replacing $M > 0$ by $N < 0$ and $f(x) > M$ by $f(x) < N$. In this case, if we consider any horizontal line $y = N$ (with N negative), the graph of f lies *below* this line whenever x is in a suitable interval $(a - \delta, a + \delta)$ and $x \neq a$.

EXERCISES 2.4

Exer. 1–10: For the given $f(x)$, express each of the following limits as ∞ , $-\infty$, or DNE (Does Not Exist):

(a) $\lim_{x \rightarrow a^-} f(x)$ (b) $\lim_{x \rightarrow a^+} f(x)$ (c) $\lim_{x \rightarrow a} f(x)$

1 $f(x) = \frac{5}{x-4}$; $a = 4$

2 $f(x) = \frac{5}{4-x}$; $a = 4$

3 $f(x) = \frac{8}{(2x+5)^3}$; $a = -\frac{5}{2}$

4 $f(x) = \frac{-4}{7x+3}$; $a = -\frac{3}{7}$

5 $f(x) = \frac{3x}{(x+8)^2}$; $a = -8$

6 $f(x) = \frac{3x^2}{(2x-9)^2}$; $a = \frac{9}{2}$

7 $f(x) = \frac{2x^2}{x^2-x-2}$; $a = -1$

8 $f(x) = \frac{4x}{x^2-4x+3}$; $a = 1$

9 $f(x) = \frac{1}{x(x-3)^2}$; $a = 3$

10 $f(x) = \frac{-1}{(x+1)^2}$; $a = -1$

Exer. 11–24: Find the limit, if it exists.

$$11 \lim_{x \rightarrow \infty} \frac{5x^2 - 3x + 1}{2x^2 + 4x - 7}$$

$$12 \lim_{x \rightarrow \infty} \frac{3x^3 - x + 1}{6x^3 + 2x^2 - 7}$$

$$13 \lim_{x \rightarrow -\infty} \frac{4 - 7x}{2 + 3x}$$

$$14 \lim_{x \rightarrow -\infty} \frac{(3x + 4)(x - 1)}{(2x + 7)(x + 2)}$$

$$15 \lim_{x \rightarrow -\infty} \frac{2x^2 - 3}{4x^3 + 5x}$$

$$16 \lim_{x \rightarrow \infty} \frac{2x^2 - x + 3}{x^3 + 1}$$

$$17 \lim_{x \rightarrow \infty} \frac{-x^3 + 2x}{2x^2 - 3}$$

$$18 \lim_{x \rightarrow -\infty} \frac{x^2 + 2}{x - 1}$$

$$19 \lim_{x \rightarrow -\infty} \frac{2 - x^2}{x + 3}$$

$$20 \lim_{x \rightarrow \infty} \frac{3x^4 + x + 1}{x^2 - 5}$$

$$21 \lim_{x \rightarrow \infty} \sqrt[3]{\frac{8 + x^2}{x(x + 1)}}$$

$$22 \lim_{x \rightarrow -\infty} \frac{4x - 3}{\sqrt{x^2 + 1}}$$

$$23 \lim_{x \rightarrow \infty} \sin x$$

$$24 \lim_{x \rightarrow \infty} \cos x$$

Exer. 25–26: Investigate the limit by letting $x = 10^n$ for $n = 1, 2, 3$, and 4.

$$25 \lim_{x \rightarrow \infty} \frac{1}{x} \tan \left(\frac{\pi}{2} - \frac{1}{x} \right)$$

$$26 \lim_{x \rightarrow \infty} \sqrt{x} \sin \frac{1}{x}$$

Exer. 27–36: Find the vertical and horizontal asymptotes for the graph of f .

$$27 f(x) = \frac{1}{x^2 - 4}$$

$$28 f(x) = \frac{5x}{4 - x^2}$$

$$29 f(x) = \frac{2x^2}{x^2 + 1}$$

$$30 f(x) = \frac{3x}{x^2 + 1}$$

$$31 f(x) = \frac{1}{x^3 + x^2 - 6x}$$

$$32 f(x) = \frac{x^2 - x}{16 - x^2}$$

$$33 f(x) = \frac{x^2 + 3x + 2}{x^2 + 2x - 3}$$

$$34 f(x) = \frac{x^2 - 5x}{x^2 - 25}$$

$$35 f(x) = \frac{x + 4}{x^2 - 16}$$

$$36 f(x) = \frac{\sqrt[3]{16 - x^2}}{4 - x}$$

Exer. 37–40: A function f satisfies the given conditions. Sketch a possible graph for f , assuming that it does not cross a horizontal asymptote.

$$37 \lim_{x \rightarrow -\infty} f(x) = 1; \quad \lim_{x \rightarrow \infty} f(x) = 1;$$

$$\lim_{x \rightarrow 3^-} f(x) = -\infty; \quad \lim_{x \rightarrow 3^+} f(x) = \infty$$

$$38 \lim_{x \rightarrow -\infty} f(x) = -1; \quad \lim_{x \rightarrow \infty} f(x) = -1;$$

$$\lim_{x \rightarrow 2^-} f(x) = \infty; \quad \lim_{x \rightarrow 2^+} f(x) = -\infty$$

$$39 \lim_{x \rightarrow -\infty} f(x) = -2; \quad \lim_{x \rightarrow \infty} f(x) = -2;$$

$$\lim_{x \rightarrow 3^-} f(x) = \infty; \quad \lim_{x \rightarrow 3^+} f(x) = -\infty;$$

$$\lim_{x \rightarrow -1^-} f(x) = -\infty; \quad \lim_{x \rightarrow -1^+} f(x) = \infty$$

$$40 \lim_{x \rightarrow -\infty} f(x) = 3; \quad \lim_{x \rightarrow \infty} f(x) = 3;$$

$$\lim_{x \rightarrow 1^-} f(x) = \infty; \quad \lim_{x \rightarrow 1^+} f(x) = -\infty;$$

$$\lim_{x \rightarrow -2^-} f(x) = -\infty; \quad \lim_{x \rightarrow -2^+} f(x) = \infty$$

41 Salt water of concentration 0.1 pound of salt per gallon flows into a large tank that initially contains 50 gallons of pure water.

(a) If the flow rate of salt water into the tank is 5 gallons per minute, find the volume $V(t)$ of water and the amount $A(t)$ of salt in the tank after t minutes.

(b) Find a formula for the salt concentration $c(t)$ (in lb/gal) after t minutes.

(c) What happens to $c(t)$ over a long period of time?

42 An important problem in fishery science is predicting next year's adult breeding population R (the recruits) from the number S that are presently spawning. For some species (such as North Sea herring), the relationship between R and S is given by $R = aS/(S + b)$, where a and b are positive constants. What happens as the number of spawners increases?

2.5 CONTINUOUS FUNCTIONS

In everyday usage we say that time is continuous, since it proceeds in an uninterrupted manner. On any given day, time does not jump from 1:00 P.M. to 1:01 P.M., leaving a gap of one minute. If an object is dropped from a hot air balloon, we regard its subsequent motion as continuous. If the initial altitude is 500 feet above ground, the object passes through every altitude between 500 and 0 feet before it hits the ground.